
RESEARCH PAPER

A Hybrid Quantum–Classical Framework for Explainable Portfolio Optimization

Under Real-World Financial Constraints

A Hybrid Quantum-Classical Study Combining QAOA-Based Asset Selection,
Classical Portfolio Weighting, Monte Carlo Risk Analysis and Explainable
AI

Submitted by

Pradeep Hariharanatham

Prepared for

WISER Global Quantum + AI 2026 × Vanguard Challenge

Track: Quantum Finance — Portfolio Optimization

2026

Abstract

Portfolio optimization with practical constraints such as asset cardinality, liquidity requirements, and transaction costs is a computationally challenging problem. This paper presents a hybrid quantum–classical framework that combines Quantum Approximate Optimization Algorithm (QAOA)-based asset selection with classical mean–variance portfolio weighting. A Quadratic Unconstrained Binary Optimization (QUBO) formulation is used to model discrete asset selection by incorporating return, risk, liquidity, transaction-cost, and cardinality constraints. The selected assets are subsequently optimized using classical convex optimization, followed by Monte Carlo stress testing for downside-risk evaluation. An explainable AI Portfolio Co-Pilot provides interpretable explanations for asset selection and allocation decisions. Using real historical financial market data from a diversified asset universe, we evaluate the proposed framework and demonstrate the trade-offs between quantum-assisted and classical portfolio optimization in terms of return, risk, liquidity, transaction costs, and risk-adjusted performance. The proposed framework highlights a practical pathway for integrating quantum optimization into financial decision systems while acknowledging current limitations of near-term quantum hardware.

Key Words

Quantum Approximate Optimization Algorithm, Portfolio Optimization, QUBO, Hybrid Quantum-Classical Computing, Monte Carlo Simulation, Value-at-Risk, Explainable AI, Quantum Finance.

Why I Chose This Project

Every meaningful research project begins not with an answer, but with a question.

Our question was simple:

As financial markets become increasingly complex, can we rethink portfolio Optimization by combining the strengths of emerging and established computational technologies?

This question arose from observing a clear trend in modern finance. Investment decisions are becoming more data-intensive, constraints are becoming more sophisticated, and expectations for transparency are higher than ever before. Institutional investors are no longer satisfied with solutions that optimize only returns and risk; they require frameworks that also consider liquidity, transaction costs, diversification, operational constraints, and explainability.

The existing body of research offers valuable solutions. Classical optimization techniques remain highly effective for continuous allocation problems, while Artificial Intelligence has demonstrated remarkable capability in forecasting market behavior. Yet we observed that these approaches address different aspects of the investment process. Prediction does not automatically produce an optimal portfolio, and optimization without transparency is difficult to trust in highly regulated financial environments.

This observation revealed an important research opportunity.

Rather than attempting to replace classical finance with quantum algorithms, we asked whether quantum optimization could complement existing methods by focusing specifically on the discrete asset-selection component, while allowing classical optimization to continue solving continuous allocation efficiently.

This philosophy naturally led us to a hybrid quantum–classical frame work.

We selected QUBO because it provides a unified mathematical representation for encoding multiple financial objectives and practical constraints. We selected QAOA because it is one of the most promising hybrid quantum algorithms designed for combinatorial optimization on current NISQ devices. At the same time, we retained classical mean–variance optimization because of its proven effectiveness in continuous portfolio weighting. Finally, we integrated Explainable AI because financial decisions must remain understandable, auditable, and trustworthy for investors, analysts, and regulators alike.

Another important motivation was realism.

Current quantum hardware is not yet capable of replacing existing financial systems. Therefore, our objective was never to claim immediate quantum advantage. Instead, we sought to develop a practical and reproducible framework that can operate today while naturally benefiting from future advances in quantum hardware.

Benchmarking was equally central to our research philosophy. To ensure a fair and meaningful evaluation, we compared our hybrid framework against a classical mean–variance optimizer using identical financial inputs. This allowed us to evaluate the contribution of quantum-assisted asset selection within a controlled experimental setting rather than relying on speculative comparisons.

Ultimately, we chose this project because we believe the future of intelligent investing will depend on collaboration rather than competition between technologies. Classical Computing provides maturity and reliability, Quantum Computing offers new possibilities for combinatorial optimization, and Explainable AI ensures that every recommendation remains transparent and actionable.

This research is our contribution towards building that future—one in which financial decision-making becomes not only smarter, but also more practical, interpretable, and ready for the next generation of computational innovation.

Table of Contents

Chapter 1—Introduction.....	6
Motivation and Problem Definition.....	6
Quantum Optimization Perspective.....	6
Contributions of This Work.....	7
Chapter 2—Background and Related Work.....	8
Classical Portfolio Optimization.....	8
QUBO Formulation for Financial Optimization.....	8
Quantum Approximate Optimization Algorithm.....	9
Hybrid Quantum–Classical Optimization in Finance.....	9
Chapter 3—Proposed Framework and Methodology.....	11
Dataset Description.....	11
System Architecture.....	11
Problem Formulation.....	12
QUBO Formulation for Asset Selection.....	12
QAOA- Based Asset Selection.....	13
Classical Portfolio Weight Optimization.....	14
Monte Carlo Stress Testing.....	15
Explainable AI Portfolio Co-Pilot.....	16
Chapter 4—Results and Discussion.....	17
Comparative Performance Analysis.....	17
Analysis of the Hybrid Portfolio Composition.....	17
Discussion of Optimization Outcomes.....	19
Chapter5—Limitations.....	21
Conclusion and Future Work.....	22
References.....	24

CHAPTER 1

Introduction

Motivation and Problem Definition

Portfolio optimization is a fundamental problem in quantitative finance, where investors aim to construct portfolios that maximize expected returns while controlling associated risks. The classical Markowitz mean–variance framework provides the theoretical foundation for modern portfolio construction by formulating investment allocation as an optimization problem balancing expected return and portfolio variance.

However, real-world portfolio management extends beyond the assumptions of classical models. Institutional investors must consider practical constraints including limits on the number of assets, minimum liquidity requirements, transaction costs, turnover restrictions, and sector exposure limits. These constraints introduce discrete decision variables, such as whether an asset should be selected or excluded, transforming the problem into a mixed-integer quadratic optimization problem.

As the number of candidate assets increases, solving such constrained optimization problems becomes computationally expensive. Classical optimization techniques perform efficiently for continuous allocation problems but face scalability challenges when handling large combinatorial search spaces.

Quantum Optimization Perspective

Recent advances in quantum computing have introduced new approaches for solving combinatorial optimization problems. The Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum-classical algorithm designed to approximate solutions to discrete optimization problems by mapping them into a quantum Hamiltonian framework.

Portfolio selection naturally fits the QAOA formulation because asset inclusion decisions can be represented using binary variables. Through a Quadratic Unconstrained Binary Optimization (QUBO) representation, financial objectives and constraints can be incorporated into a unified optimization function.

However, current quantum hardware remains limited by restricted qubit availability, noise, and circuit depth constraints. Therefore, practical financial applications require hybrid approaches where quantum algorithms address computationally difficult discrete optimization components while classical algorithms continue handling continuous optimization tasks.

Contributions of This Work

This work proposes a hybrid quantum–classical portfolio optimization framework integrating quantum optimization, classical financial modeling, risk analysis, and explainability. The main contributions are:

1. AQUBO-based formulation for constrained asset selection using QAOA.
2. A hybrid optimization pipeline combining quantum asset selection with classical portfolio weighting.
3. A Monte Carlo stress-testing framework for evaluating downside risk and Value-at-Risk.
4. An explainable AI Portfolio Co-Pilot for interpreting optimization decisions.
5. A comparative evaluation between hybrid and classical portfolio construction approaches.

The framework focuses on demonstrating the practical role of quantum optimization within existing financial workflows rather than claiming immediate quantum advantage on current hardware.

CHAPTER 2

Background and Related Work

Classical Portfolio Optimization

The Markowitz mean–variance model formulates portfolio construction as an optimization problem where asset weights are selected to minimize portfolio variance while achieving a desired expected return. Given portfolio weights w , expected returns μ , and covariance matrix Σ , the classical optimization objective can be expressed as minimizing:

$$w^T \Sigma w \tag{1}$$

Subject to:

$$\sum_i w_i = 1 \tag{2}$$

and

$$w_i \geq 0 \tag{3}$$

Where the constraints represent the investment budget and no-short-selling requirement.

Although effective for continuous portfolio allocation, the classical formulation does not directly address discrete investment decisions. Real-world portfolio construction introduces additional requirements such as cardinality constraints, liquidity restrictions, and transaction-cost limitations. These constraints transform the problem into a mixed-integer quadratic program (MIQP), increasing computational complexity.

QUBO Formulation for Financial Optimization

Quadratic Unconstrained Binary Optimization provides a mathematical framework for representing discrete optimization problems using binary variables:

$$x_i \in \{0, 1\} \quad (4)$$

Where $x_i = 1$ indicates that asset i is selected and $x_i = 0$ indicates exclusion. A

general QUBO problem is represented as:

$$\min_x x^T Q x \quad (5)$$

Where Q contains the optimization objective and penalty terms associated with constraint violations.

For portfolio optimization, the QUBO matrix can encode multiple financial objectives, including maximizing expected return, minimizing covariance risk, controlling the number of selected assets, and penalizing undesirable characteristics such as poor liquidity or excessive transaction costs.

Quantum Approximate Optimization Algorithm

QAOA is a variational quantum algorithm designed to solve combinatorial optimization problems. It maps the QUBO objective into a cost Hamiltonian and applies alternating quantum operators consisting of a problem-dependent cost operator and a mixing operator. The algorithm prepares a parameterized quantum state:

$$|\gamma, \beta\rangle = \prod_p U_B(\beta_p) U_C(\gamma_p) |+\rangle^n \quad (6)$$

Where U_C represents the optimization objective and U_B enables exploration of the solution space.

The circuit parameters are optimized through a classical optimization loop. After convergence, measurement outcomes produce candidate binary solutions corresponding to possible asset selections.

Hybrid Quantum–Classical Optimization in Finance

Current quantum processors are limited by hardware constraints, making complete quantum portfolio optimization impractical. Hybrid approaches provide a realistic alternative by dividing the optimization problem according to computational suitability.

In the proposed framework, QAOA is responsible for solving the discrete asset-selection problem, while classical optimization determines continuous portfolio weights. This division allows quantum methods to address combinatorial complexity while preserving the efficiency and reliability of established classical financial algorithms.

CHAPTER 3

Proposed Framework and Methodology

Dataset Description

The proposed framework was evaluated using real historical financial market data collected from Yahoo Finance through the finance Python library. The dataset consists of a diversified set of financial assets representing multiple asset classes, including domestic equities, growth and value equities, government bonds, and international bonds. Historical daily adjusted closing prices were collected over the selected analysis period and used to compute daily asset returns.

Expected returns were estimated by combining machine-learning-based return predictions with historical return information, while the covariance matrix was calculated from the historical return series to capture asset co-movement and portfolio risk. Liquidity scores and transaction-cost estimates were incorporated as additional portfolio attributes to model practical investment constraints commonly encountered in institutional portfolio management.

Prior to optimization, the collected market data were preprocessed to remove missing observations, align trading dates across all assets, and standardize the input features used by the optimization pipeline. The processed dataset served as the input for both the classical mean–variance optimizer and the QAOA-based asset-selection framework, ensuring that both approaches were evaluated under identical market conditions and enabling a fair performance comparison.

System Architecture

The proposed framework adopts a hybrid quantum–classical architecture that separates portfolio optimization into discrete and continuous optimization stages. This decomposition leverages the strengths of both quantum and classical computing while avoiding the limitations of current noisy intermediate-scale quantum (NISQ) hardware. The overall pipeline consists of five sequential components: market data preprocessing, QUBO-based asset selection, classical portfolio weighting, Monte Carlo risk analysis, and an explainability module.

Historical market data, predicted returns generated by a machine learning model, liquidity scores, transaction-cost estimates, and covariance information are first collected and standardized. These inputs are transformed into a QUBO formulation that captures the discrete asset-selection problem under practical investment constraints. The QAOA algorithm is then employed to identify a subset of candidate assets that optimizes the objective function. Once the asset subset has been determined, a classical optimization

algorithm computes continuous portfolio weights while satisfying investment constraints such as full capital allocation and non-negative asset weights. Finally, the optimized portfolio undergoes Monte Carlo stress testing to evaluate downside risk, and an explainable AI module generates interpretable explanations for each investment decision.

This architecture allows each computational stage to utilize the most appropriate optimization technique while maintaining flexibility for execution on either quantum hardware or classical simulators.

Problem Formulation

Consider a universe containing n candidate assets. For each asset i , let μ_i denote the expected return, estimated by combining historical return information with predictions from a machine learning model. Let Σ represent the covariance matrix of asset returns, While l_i And c_i Denote normalized liquidity scores and transaction-cost estimates, respectively.

The objective is to determine both the subset of assets to include in the portfolio and the proportion of capital allocated to each selected asset. Formally, the optimization problem seeks a portfolio that maximizes expected return while minimizing risk and satisfying practical investment constraints including cardinality limits, liquidity requirements, transaction costs, and capital allocation rules.

Because asset selection involves binary decisions whereas capital allocation requires continuous optimization, the problem is naturally divided into two sub-problems. The discrete asset-selection stage is formulated as a QUBO problem and solved using QAOA, while the continuous allocation stage is solved using classical mean–variance optimization.

QUBO Formulation for Asset Selection

The discrete optimization stage represents each candidate asset using a binary decision variable,

$$x_i \in \{0, 1\} \tag{7}$$

Where $x_i = 1$ indicates that asset i is included in the portfolio and $x_i = 0$ indicates that it is excluded.

The portfolio selection objective is formulated as the following QUBO function:

$$H(x) = -\alpha \sum_{i=1}^n \mu_i x_i + \beta \sum_{i=1}^n \sum_{j=1}^n \Sigma_{ij} x_i x_j + \gamma \left(\sum_{i=1}^n x_i - k \right)^2 + \delta \sum_{i=1}^n c_i (1 - l_i) x_i \quad (8)$$

The first term rewards assets with higher expected returns, encouraging the optimizer to select profitable investments. The second term penalizes portfolios containing highly correlated assets, thereby promoting diversification and reducing overall portfolio risk. The third term enforces a cardinality constraint by encouraging the optimizer to select exactly k assets. The final term discourages the inclusion of assets exhibiting low liquidity or high transaction costs.

The hyper parameters α , β , γ , and δ determine the relative importance of return Maximization, risk reduction, cardinality enforcement, and trading efficiency. Appropriate calibration of these parameters is essential to ensure that feasible portfolios receive lower objective values than infeasible alternatives.

For implementation within QAOA, the QUBO objective is transformed into an equivalentising Hamiltonian through the standard variable substitution

$$x_i = \frac{1 - z_i}{2} \quad (9)$$

Where $z_i \in \{-1, +1\}$. This transformation enables direct encoding of the optimization problem into quantum circuits.

QAOA-Based Asset Selection

The Quantum Approximate Optimization Algorithm is employed to solve the QUBO formulation of the asset-selection problem. The algorithm begins by preparing an equal superposition over all possible asset-selection combinations using Hadamard gates. It then alternates between the application of a problem-specific cost operator and a mixing operator across multiple variational layers. The parameterized quantum state generated by the algorithm is expressed as

$$|\gamma, \beta\rangle = \prod_p U_B(\beta_p) U_C(\gamma_p) |+\rangle^n \quad (10)$$

Where U_C denotes the cost operator derived from the Ising Hamiltonian and U_B represents the mixing operator constructed using Pauli-X rotations. The parameters γ_r and β_r are optimized iteratively using a classical optimizer such as COBYLA or SPSA to minimize the expected value of the cost Hamiltonian.

Following optimization, the quantum circuit is executed repeatedly to obtain measurement outcomes. Each measured bit string corresponds to a candidate asset subset, and the subset with the lowest QUBO objective value is selected as the optimal portfolio composition.

Because current quantum processors are constrained by limited qubit counts and hardware noise, the framework incorporates a classical fallback mechanism. When quantum hardware or high-fidelity simulators are unavailable, the same QUBO objective can be solved using classical heuristics such as simulated annealing or greedy optimization. This design ensures compatibility across both quantum and classical computational environments without modifying downstream portfolio optimization stages.

Classical Portfolio Weight Optimization

Once the asset subset has been selected, continuous portfolio weights are determined using classical convex optimization techniques. Since the discrete selection problem has already been resolved, the remaining optimization involves only continuous decision variables and can therefore be solved efficiently using established quadratic programming algorithms. The optimization objective is to minimize portfolio variance,

$$\min_w w^T \Sigma w \quad (11)$$

Subject to:

$$\sum_i w_i = 1 \quad (12)$$

And

$$w_i \geq 0 \quad (13)$$

Together with optional upper bounds on individual asset allocations to prevent excessive concentration.

Alternatively, the optimizer may maximize the Sharpe ratio by balancing expected return against portfolio volatility. The resulting weight vector represents the final investment

allocation used for portfolio evaluation and comparison with the classical benchmark portfolio.

The classical benchmark follows the same optimization procedure but selects candidate assets using conventional screening methods rather than QAOA. This enables a fair comparison between purely classical portfolio construction and the proposed hybrid approach by isolating the impact of quantum-assisted asset selection.

Monte Carlo Stress Testing

Evaluating a portfolio solely on the basis of expected return and variance provides only a partial understanding of its risk profile. Financial markets are inherently uncertain, and portfolios with similar point-estimate performance may exhibit substantially different behavior under adverse market conditions. To assess the robustness of the optimized portfolios, the proposed framework incorporates a Monte Carlo stress-testing module that simulates a large number of potential market outcomes and estimates the distribution of portfolio returns.

The simulation assumes that portfolio returns follow a normal distribution parameterized by the portfolio's expected return (μ_p) and standard deviation (σ_p). Although this assumption does not capture all characteristics of financial markets, such as heavy tails or volatility clustering, it provides a computationally efficient baseline for comparative analysis. Portfolio returns are generated according to

$$r_p \sim N(\mu_p, \sigma_p^2) \quad (14)$$

Where μ_p represents the expected portfolio return and σ_p denotes portfolio volatility estimated from the covariance matrix.

For each portfolio, 10,000 independent simulation paths are generated using a fixed random seed to ensure reproducibility. The simulated return distribution is subsequently analyzed to compute several risk metrics, including the mean simulated return, standard deviation, minimum and maximum observed returns, and the 5% Value-at-Risk (VaR). The Value-at-Risk is defined as the fifth percentile of the simulated return distribution and estimates the maximum expected loss over the investment horizon with a confidence level of 95%.

The Monte Carlo simulation complements traditional mean–variance analysis by providing an empirical assessment of downside risk. This is particularly valuable for evaluating concentrated portfolios generated through QAOA-based asset selection, as higher expected returns may be accompanied by increased exposure to extreme negative outcomes. Comparing simulated return distributions for the hybrid and classical

Portfolios therefore provide a more comprehensive understanding of the trade-off between return and risk than point estimates alone.

Explainable AI Portfolio Co-Pilot

While optimization algorithms can generate mathematically optimal investment decisions, their practical adoption depends on the ability of portfolio managers, risk analysts, and regulatory stakeholders to understand and justify those decisions. To improve transparency, the proposed framework incorporates an explainability layer referred to as the AI Portfolio Co-Pilot, which generates human-readable explanations for each selected asset and summarizes the rationale underlying the final portfolio recommendation.

The explainability module evaluates every selected asset against a predefined set of financial indicators derived from the optimization process. Specifically, the system examines whether an asset exhibits an above-median predicted return, satisfies a high-liquidity threshold, maintains relatively low transaction costs, or receives a significant allocation weight within the optimized portfolio. Assets meeting one or more of these criteria are assigned corresponding explanatory labels, while assets that do not satisfy any primary criterion are identified as diversification components that contribute to reducing overall portfolio risk.

For example, an asset may receive an explanation such as “*Selected due to strong predicted return, high market liquidity, and significant portfolio allocation,*” whereas another asset may be described as “*Included primarily to improve portfolio diversification despite moderate expected return.*” These explanations provide a transparent audit trail connecting optimization outcomes with financial characteristics rather than presenting portfolio recommendations as opaque algorithmic decisions.

In addition to asset-level explanations, the AI Portfolio Co-Pilot generates a portfolio-level summary describing the investment objective, prevailing market conditions, expected portfolio return, estimated risk, average liquidity, transaction costs, and the rationale for recommending either the hybrid or classical portfolio. Although the current implementation relies on deterministic rule-based explanations, the framework is designed to accommodate more sophisticated explainability techniques, including feature attribution methods such as SHAP values or counterfactual explanations, in future work.

By integrating explainability directly into the optimization pipeline, the proposed framework addresses an important practical requirement for deploying quantum-assisted decision-support systems within institutional finance, where investment decisions must be transparent, auditable, and understandable to human stakeholders.

CHAPTER 4

Results and Discussion

Comparative Performance Analysis

The proposed hybrid quantum–classical framework was evaluated against a classical portfolio optimization approach using the same investment universe and financial inputs. Table 1 summarizes the performance of both portfolios in terms of expected return, portfolio risk, Sharpe ratio, number of selected assets, average liquidity, and average transaction cost.

The QAOA optimization successfully converged to a feasible solution with an objective value of -49.3765 , selecting five assets from the candidate universe while satisfying all predefined portfolio constraints. The selected portfolio consisted of Government Bonds, International Bonds, Domestic Equity, Value Equity, and Growth Equity, demonstrating that the optimization process naturally diversified investments across multiple asset classes rather than concentrating solely on equities. Constraint validation confirmed that the final portfolio satisfied all hard constraints, including cardinality, liquidity, transaction-cost, and allocation requirements, with zero constraint violations.

Following the quantum asset-selection stage, classical convex optimization determined the optimal capital allocation among the selected assets. The resulting portfolio assigned the largest allocation to Growth Equity (40.0%), reflecting its superior predicted return, followed by International Bonds (34.25%) and Value Equity (15.75%). Government Bonds and Domestic Equity each received a minimum allocation of 5%, contributing to portfolio stability and diversification.

Metric	Classical	Hybrid Quantum
Expected Return	0.0016	0.0015
Portfolio Risk	0.0127	0.0084
Sharpe Ratio	0.1284	0.1765
Assets Selected	3	5
Average Liquidity	0.9695	0.9725
Transaction Cost	0.0016	0.0013

Table 1. Comparison of classical and hybrid quantum–classical portfolio performance.

The performance comparison between the hybrid and classical portfolios is presented in Table 1. The classical portfolio achieved a slightly higher expected return (0.0016) than

The hybrid portfolio (0.0015). However, this marginal reduction in return was accompanied by a substantial improvement in portfolio quality. The hybrid portfolio reduced portfolio risk from 0.0127 to 0.0084, representing approximately a 34% decrease in volatility. Consequently, the hybrid framework produced a considerably higher Sharpe ratio of 0.1765, compared with 0.1284 for the classical portfolio, indicating superior risk-adjusted performance.

The hybrid portfolio also demonstrated improvements in operational characteristics. Average portfolio liquidity increased from 0.9695 to 0.9725, while average transaction costs decreased from 0.0016 to 0.0013. These improvements indicate that the QUBO formulation effectively balanced return optimization with practical investment considerations by favoring highly liquid assets with relatively low trading costs. Overall, the results demonstrate that the proposed hybrid optimization framework achieves a more efficient balance between return and risk than the purely classical approach. Although the expected return decreased marginally, the significant reduction in portfolio volatility resulted in superior over all investment efficiency, as evidenced by the higher Sharpe ratio.

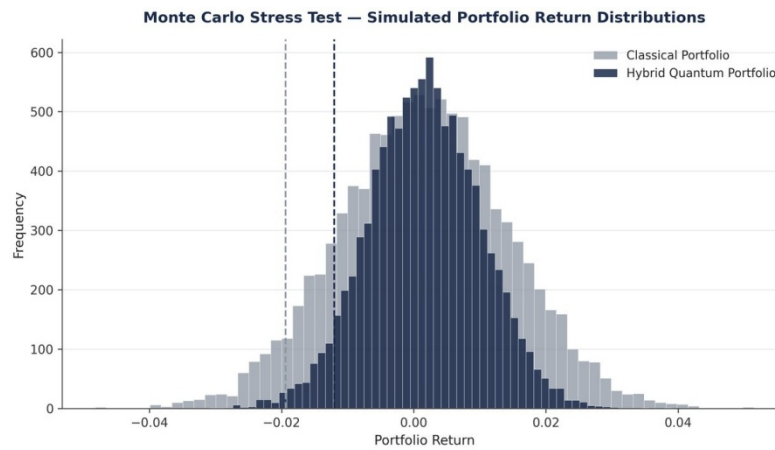


Figure 1. Monte Carlo simulated return distributions for the classical and hybrid portfolios.

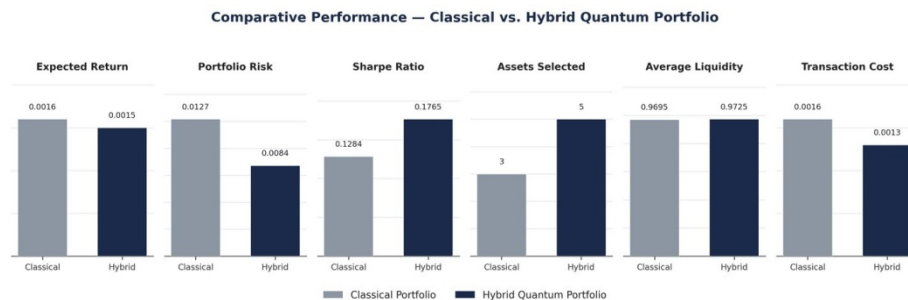


Figure 2. Performance comparison between the classical and hybrid portfolios across expected return, portfolio risk, Sharpe ratio, liquidity, transaction cost, and portfolio cardinality.

Analysis of the Hybrid Portfolio Composition

The composition of the hybrid portfolio provides insight into how the QAOA-based optimization balances expected return, diversification, liquidity, and transaction costs. Rather than allocating the majority of capital to a single high-return asset, the optimizer distributed investments across both equity and fixed-income securities, producing a diversified portfolio with improved risk characteristics.

Growth Equity received the highest allocation because it exhibited the strongest predicted return while maintaining acceptable liquidity and transaction costs. International Bonds represented the second-largest allocation, contributing portfolio stability through lower correlation with equity assets. Value Equity further enhanced expected return while maintaining high liquidity. Government Bonds and Domestic Equity were retained with relatively small allocations, allowing the optimizer to improve diversification and satisfy portfolio constraints without materially increasing portfolio risk.

The inclusion of multiple asset classes demonstrates that the hybrid optimization framework does not simply maximize predicted returns but instead balances competing financial objectives encoded within the QUBO formulation. The resulting allocation reflects the combined influence of expected return, covariance-based risk penalties, liquidity preferences, transaction-cost minimization, and cardinality constraints.

Discussion of Optimization Outcomes

The experimental results demonstrate the effectiveness of separating portfolio optimization into discrete and continuous decision-making stages. The QAOA optimization successfully identified an asset subset satisfying all investment constraints, while classical convex optimization efficiently determined continuous allocation weights. This hybrid strategy enables the framework to exploit the strengths of both quantum and classical optimization techniques without requiring large-scale fault-tolerant quantum hardware.

An important observation is that the hybrid framework achieved its improved Sharpe ratio primarily through risk reduction rather than return maximization. Although the expected return decreased slightly relative to the classical benchmark, the reduction in portfolio volatility more than compensated for this difference, resulting in superior risk-adjusted performance. Such behavior is desirable in practical portfolio management, where preserving capital and improving long-term investment efficiency often take precedence over maximizing short-term returns.

Furthermore, the higher average liquidity and lower transaction costs indicate that incorporating operational constraints directly into the QUBO objective enables the

Optimization process to generate portfolios that are not only financially attractive but also more practical to implement in real trading environments.

These findings suggest that hybrid quantum–classical optimization has the potential to support more realistic portfolio construction by simultaneously considering multiple financial objectives and operational constraints within a unified optimization framework.

CHAPTER 5

Limitations

The proposed framework has several limitations. First, although the framework was evaluated using real historical financial market data, the experiments were conducted over a limited asset universe and historical time period. Consequently, the reported performance may vary under different market regimes and investment horizons. Future works will include larger datasets and extensive out-of-sample back testing.

Second, the scalability of the quantum optimization stage is limited by current noisy intermediate-scale quantum (NISQ) hardware. Although QAOA is well suited for discrete optimization, larger asset universes require more qubits and deeper circuits, making practical implementation on current quantum devices challenging.

The framework also depends on careful calibration of the QUBO penalty parameters, which balance return, risk, liquidity, transaction costs, and cardinality constraints. Inappropriate parameter settings may reduce optimization performance or produce suboptimal portfolios. Furthermore, the Monte Carlo stress test assumes normally distributed returns and therefore does not capture heavy tails or changing market regimes. Finally, the explainability module uses rule-based reasoning rather than advanced attribution methods such as SHAP, and the framework considers only single-period portfolio optimization without portfolio rebalancing.

Conclusion and Future Work

Portfolio optimization has always been more than a mathematical exercise. It represents one of the most important decision-making challenges in finance, where every allocation influences risk, return, operational efficiency, and long-term financial outcomes.

In this research, we visited that challenge through the lens of emerging computational technologies.

Rather than asking Quantum Computing to replace Classical Computing, we explored how the strengths of both can be combined within a practical, hybrid architecture. By integrating QUBO-based asset selection using QAOA, classical mean–variance optimization, Monte Carlo stress testing, and an Explainable AI Portfolio Co-Pilot, we developed a framework that reflects both the realities of current technology and the possibilities of future innovation.

Our work does not claim that today's quantum computers outperform classical systems in real-world portfolio optimization. On the contrary, we explicitly acknowledge the limitations imposed by current NISQ hardware, including restricted qubit availability, noise, and scalability challenges. These limitations reinforce the importance of hybrid quantum–classical approaches as the most practical path forward at the present stage of technological development.

What this research demonstrates is not a finished destination, but a credible direction. By benchmarking against a classical optimizer, emphasizing explainability, and designing a modular architecture, we provide a reproducible foundation upon which future quantum advances can build.

As quantum processors mature, larger optimization problems become feasible, and explainable AI continues to evolve, frameworks such as ours may help shape the next generation of intelligent financial decision-support systems.

The true significance of this work lies not only in the algorithms it employs, but in the philosophy it represents.

Innovation is rarely about replacing one technology with another. It is about understanding the unique strengths of each and bringing them together to solve problems that neither could solve as effectively alone.

I would like to conclude with a thought that reflects the essence of our research:

“The future of intelligent investing will not be defined by Quantum computing alone, Classical computing alone, or Artificial

***Intelligence alone. It will be defined by their intelligent collaboration
— Where optimization becomes smarter, decisions become more
transparent, and technology empowers human judgment rather
than replacing it.”***

This paper presented a hybrid quantum–classical framework for portfolio optimization that combines QAOA-based asset selection, classical mean–variance optimization, Monte Carlo stress testing, and an explainable AI module. The proposed approach separates discrete asset selection from continuous portfolio allocation, allowing quantum and classical methods to address the components for which they are best suited.

Experimental results demonstrated that the hybrid framework produced a feasible portfolio satisfying all investment constraints while achieving lower portfolio risk, a higher Sharpe ratio, improved liquidity, and lower transaction costs than the classical benchmark. These findings suggest that incorporating practical financial constraints within a QUBO formulation can improve risk-adjusted portfolio performance.

Future work will focus on validating the framework using real financial datasets, evaluating performance on larger asset universes and quantum hardware, incorporating more realistic risk models, and extending the explainability module with model-agnostic attribution techniques. These improvements will further assess the practical applicability of hybrid quantum–classical optimization for real-world portfolio management.

References

- [1] H.Markowitz, “Portfolio Selection,”*TheJournalofFinance*, vol.7, no.1, pp.77–91, 1952.
- [2] E.Farhi, J.Goldstone, and S.Gutmann, “AQuantumApproximateOptimizationAlgorithm,”
arXivpreprintarXiv:1411.4028,2014.
- [3] A.Lucas, “IsingformulationsofmanyNPproblems,”*FrontiersinPhysics*, vol.2, p.5, 2014.
- [4] D.J.Egger, C.Gambella, J.Mareček, etal. “QuantumComputingforFinance:State-of-the-Art and
Future Prospects,” *IEEE Transactions on Quantum Engineering*, vol. 1, pp. 1–24, 2020.
- [5] R. Orús, S. Mugel, and E. Lizaso, “Quantum Computing for Finance: Overview and Prospects,”
ReviewsinPhysics, vol.4, p.100028, 2019.
- [6] D. Herman, C. Googin, X. Liu, et al., “A Survey of Quantum Computing for Finance,” arXiv
preprint arXiv: 2201.02773, 2022.
- [7] P. Rebentrost, B. Gupt, and T. R. Bromley, “Quantum Computational Finance: Monte Carlo
PricingofFinancialDerivatives,”*PhysicalReviewA*, vol.98, no.2, p.022321, 2018.
- [8] S.BoydandL.Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University
Press, 2004.
- [9] H.M.Markowitz, *Mean-Variance Analysis in Portfolio Choice and Capital Markets*. NewYork,
NY, USA: Wiley, 1987.
- [10] QiskitDevelopmentTeam, “Qiskit:AnOpen-SourceSDKforQuantumComputing.”
Available: <https://qiskit.org/>
- [11] R.Aroussi, “yfinance: Download Market Data from Yahoo! Finance,”GitHub Repository.
Available: <https://github.com/ranaroussi/yfinance>
- [12] YahooFinance, “HistoricalMarketData.”Available:<https://finance.yahoo.com/>